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$$f(x, y) = x^2 - 2xy + y^3 - y$$

$$\frac{\partial f}{\partial x} = 2x - 2y \Rightarrow x = y$$

$$\frac{\partial f}{\partial y} = -2x + 3y^2 - 1$$

$$\Rightarrow -2y + 3y^2 - 1 = 0$$

$$y_1 = 1$$

$$y_2 = \frac{-1}{3}$$

$$\Rightarrow \text{kritiekpunten zijn: } (1, 1) \text{ en } \left(\frac{-1}{3}, \frac{-1}{3}\right)$$

$$H(x, y) = \begin{pmatrix} 2 & -2 \\ -2 & 6y \end{pmatrix}$$

$$D = 12y - 4$$

$$H(1, 1) = 8 \Rightarrow \text{minimum want } \frac{\partial^2 f}{\partial x^2} = 2$$

$$H\left(\frac{-1}{3}, \frac{-1}{3}\right) = -8 \Rightarrow \text{zadelpunt}$$

References